

SmartImageSearch

Content-Based Image Retrieval System

Technical Project Report

Multimedia Data Management (M) — Second Cycle Degree in Computer Engineering
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Abstract

This report presents SmartImageSearch, a prototype Content-Based Image Retrieval (CBIR) system developed as part of the Multimedia Data Management course at the University of Bologna. The system retrieves images by analysing their visual content rather than relying on textual annotations, directly addressing the Semantic Gap problem that lies at the heart of modern multimedia information management.

The project covers the full engineering lifecycle: from multimedia data modelling (relational, semi-structured XML, and low-level feature descriptors) through Information Retrieval theory (Boolean model, Vector Space Model, tf-idf weighting, Precision/Recall evaluation) to a dual Python implementation comparing a classical colour-histogram pipeline against a modern deep embedding pipeline based on a pre-trained ResNet18 network.

Experimental results on a CIFAR-10 subset (4 classes, 400 images) show that the modern pipeline achieves substantially higher Precision@1 and Precision@5 than the classical approach, confirming the value of learned semantic representations over handcrafted descriptors.

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1. Introduction and Motivation

The exponential growth of digital image collections — driven by social media, e-commerce, medical imaging, and ubiquitous mobile cameras — has exposed a fundamental limitation of traditional database systems: the inability to query by visual content. Conventional search engines rely on textual metadata attached to images (filenames, tags, captions), yet such annotations are often incomplete, subjective, or simply absent. As a result, retrieving visually similar images remains a largely unsolved challenge for keyword-based systems.

SmartImageSearch addresses this problem by implementing Content-Based Image Retrieval (CBIR), a paradigm in which images are indexed and retrieved according to their intrinsic visual properties — colour distribution, texture, and shape — extracted automatically from the raw pixel data. This approach eliminates the bottleneck of manual annotation, scales to large collections, and supports the natural Query-by-Example (QBE) interaction model: the user provides a sample image and the system returns the most visually similar items from the database.

The project is developed within the framework of the Multimedia Data Management (MDM) course at the University of Bologna, which studies how modern information systems must evolve beyond the relational model to handle heterogeneous, high-dimensional, and semantically rich data objects. SmartImageSearch serves as a practical instantiation of these theoretical principles, covering the full pipeline from data modelling to implementation and quantitative evaluation.

2. Theoretical Background

2.1 Multimedia Data Management

Multimedia Data Management (MDM) emerged as a research discipline in the early 1990s in response to the proliferation of digital images, audio, and video. Unlike relational databases, which are optimised for structured, typed, atomic data, MDM systems must handle objects whose meaning is not immediately conveyed by the raw bit sequence. The discipline draws on signal processing, computer vision, machine learning, pattern analysis, and human-computer interaction.

A core taxonomic principle in MDM is the classification of data by dimensionality: zero-dimensional data (text, numeric values), one-dimensional data (audio, time series), two-dimensional data (images, graphics), and three-dimensional data (video, animation). Each category imposes distinct requirements on storage, indexing, and retrieval algorithms. Images — the focus of this project — are 2-D arrays of pixels whose semantic interpretation requires dedicated descriptors and similarity functions that go far beyond the equality predicates of SQL.

Historically, MDM systems are characterised by a tension between two retrieval paradigms: Concept-Based Retrieval, which relies on human-assigned textual annotations, and Content-Based Retrieval, which derives descriptors automatically from the signal. SmartImageSearch implements the latter, augmented by optional textual metadata to support hybrid queries.

2.2 The Semantic Gap

Definition — The Semantic Gap

The Semantic Gap is defined as the discrepancy between the low-level numerical representation of a multimedia object (pixel matrices, colour histograms, wavelet coefficients) and the high-level semantic meaning attributed to it by a human observer. It is the central engineering challenge of all CBIR systems.

Example: a photograph of a cheetah and a photograph of an eagle, both taken against a clear blue sky, may yield nearly identical colour histograms — yet they belong to entirely different semantic categories. A system operating purely on colour statistics would incorrectly rank them as highly similar.

Bridging the Semantic Gap requires progressively richer representations. The hierarchy of content description levels recognised in the MDM literature is as follows:

- **Low-level features:** Raw visual statistics: colour histograms, texture descriptors (co-occurrence matrices, Gabor filters), shape moments, edge maps. Automatically extractable but semantically weak.
- **Mid-level features:** Textual annotations, categorical tags, object-level attributes. Require human intervention or supervised learning; bridge low-level signals and semantic concepts.
- **High-level semantics:** Conceptual knowledge: scene categories, events, ontological relationships. Captured by deep neural networks and ontology-based reasoning systems such as WordNet.

Modern CBIR systems — including the deep pipeline implemented in this project — attempt to operate at the mid-to-high level by using Convolutional Neural Networks (CNNs) pre-trained on large-scale datasets such as ImageNet. The activations of deep layers implicitly encode semantic concepts, substantially reducing the gap between pixel values and human perception.

2.3 Retrieval Paradigms: Concept-Based vs. Content-Based

Aspect	Concept-Based Retrieval	Content-Based Retrieval
Mechanism	Lexical descriptions (manual or semi-automatic annotations)	Low-level feature descriptors extracted automatically from the signal
Origins	Information Retrieval (1950s)	Computer vision and signal processing (1990s)
Advantages	Intuitive textual querying; high semantic precision when annotations are accurate	Fully automatic; objective; zero annotation cost; scales to millions of items
Limitations	Unsustainable human cost; annotator subjectivity; incomplete coverage	Semantic Gap; inability to capture high-level concepts from pixels alone
Typical query	SQL / keyword search	Query-by-Example (QBE); Query-by-Sketch

Table 1 — Comparison between Concept-Based and Content-Based retrieval paradigms.

3. Multimedia Data Modelling

A robust MDM system must manage data at three distinct levels of structure, each requiring a different modelling approach. SmartImageSearch adopts a layered architecture that separates structured metadata, semi-structured annotations, and unstructured visual descriptors.

3.1 Structured Data — Relational Model

Fixed-schema metadata — image identifiers, file paths, dimensions, formats, and category labels — are stored in a relational database following the model proposed by Codd (1970). Two primary tables organise the data:

Table	Primary Key	Key Attributes
Images	ImageID (INT)	filepath, width, height, format, category, created_at
Features	ImageID (FK → Images)	feature_vector (BLOB), descriptor_type, dimensions

Table 2 — Relational schema for SmartImageSearch.

The foreign key constraint between Features.ImageID and Images.ImageID enforces referential integrity: a feature vector cannot exist without a corresponding image record. This design cleanly separates the storage of raw images (managed at the file-system level) from the storage of their compact numerical representations (managed inside the database).

3.2 Semi-Structured Data — XML Representation

Semi-structured annotations — variable-length tag lists, confidence scores, and automatically generated descriptions — are encoded in XML. Unlike the relational schema, the XML structure is self-describing: tag names carry semantic meaning and the hierarchy can vary between documents without schema violations.

```
<image id="img_001"> <metadata> <category>dog</category> <tags> <tag confidence="0.91">animal</tag> <tag confidence="0.85">outdoor</tag> </tags> </metadata> <features descriptor="color_histogram" dims="48"/> </image>
```

Listing 1 — XML document representing one image in SmartImageSearch.

3.3 Unstructured Data — Low-Level Feature Descriptors

Raw images are unstructured data: sequences of bytes with no predefined schema. They are made searchable by projecting them onto a feature space through descriptor extraction. SmartImageSearch defines two classes of descriptors:

- **Global features (Colour Histogram):** A normalised histogram of pixel intensities across the RGB colour channels. With 16 bins per channel the feature vector has 48 dimensions. Global features capture the overall colour distribution of the image but discard spatial layout information.
- **Deep embeddings (ResNet18):** A 512-dimensional embedding produced by the penultimate layer of a pre-trained ResNet18 network. These vectors encode hierarchical visual patterns — textures, parts, objects — learned from over one million training images, providing far richer semantic content than handcrafted descriptors.

4. Information Retrieval Module

SmartImageSearch incorporates a textual Information Retrieval (IR) subsystem that operates on the short text documents associated with each image (tags, labels, category names, captions). This module provides keyword-based querying and lays the foundation for future hybrid retrieval combining visual and textual signals.

4.1 Boolean Retrieval and the Inverted Index

The simplest retrieval model represents each document as a binary vector over the vocabulary V . Boolean queries combine terms with AND, OR, and NOT operators, evaluated via bitwise operations on posting lists. While semantically transparent, this approach has two key limitations: results are unranked, and the full incidence matrix is prohibitively large for realistic collections.

For a collection of $N = 10^6$ documents and $|V| = 10^5$ terms, the incidence matrix would require 10^{11} bits (approx. 100 GB) yet be 99% sparse (assuming 1 000 unique terms per document). The Inverted Index solves this by storing, for each term, only the ordered list of document IDs in which it appears (the posting list).

Construction proceeds in four phases: (1) tokenisation and pre-processing of all documents, (2) alphabetical sorting of (term, docID) pairs, (3) merging of duplicate entries and computation of term frequencies, and (4) splitting into a Dictionary (per-term statistics) and a Posting File (per-term document lists). AND queries are answered by intersecting posting lists in linear time; OR queries by merging them.

Heap's Law — Vocabulary Growth

The empirical size of the vocabulary follows Heap's Law: $V = k \cdot n^b$, where n is the total number of tokens in the corpus, and typical parameters are $b \approx 0.5$ and k in $[30, 100]$. This sub-linear growth confirms that the inverted index dictionary remains tractable even as the corpus scales to billions of documents.

Pre-processing optimisations further reduce index size: stopword removal (the 30 most frequent words cover ~30% of all tokens) reduces posting list volume by 25–30%; stemming (reducing terms to their morphological root, e.g. 'compression' → 'compress') shrinks the vocabulary by ~40% and the overall index by ~30%.

4.2 Vector Space Model and tf-idf Weighting

The Vector Space Model (VSM) overcomes the binary limitation by representing documents and queries as real-valued vectors in $\mathbb{R}^{|V|}$, where each component is the tf-idf weight of the corresponding term. Relevance becomes a geometric concept: the cosine of the angle between the query vector and each document vector.

The tf-idf weight for term t in document d is defined as:

$$w(t, d) = \text{tf}(t, d) \times \text{idf}(t)$$

where $\text{tf}(t, d)$ is the raw (or log-normalised) frequency of term t in document d , and $\text{idf}(t) = \log(N / \text{df}(t))$ measures the discriminative power of t across the N -document collection ($\text{df}(t)$ = number of documents containing t). Terms that appear in nearly every document receive very low idf, preventing them from dominating retrieval scores.

Cosine similarity between query q and document d is computed as:

$$\text{sim}(q, d) = (q \cdot d) / (|q| \cdot |d|)$$

The L2 normalisation of both vectors (division by their magnitudes) is essential: without it, longer documents would systematically outrank shorter ones regardless of topical relevance. The VSM supports partial matching — documents sharing only a subset of query terms receive non-zero scores — and naturally produces a ranked result list.

4.3 Evaluation Metrics: Precision and Recall

System effectiveness is measured using the standard Precision / Recall framework, derived from the four-region partition of the document universe:

Region	Meaning	Impact on metrics
A (Relevant $\cap$ Retrieved)	True positives — correct hits	Increases both P and R
B (Not Relevant $\cap$ Retrieved)	False positives — false drops	Decreases Precision
C (Not Relevant $\cap$ Not Retrieved)	True negatives — correct rejections	No impact
D (Relevant $\cap$ Not Retrieved)	False negatives — missed items	Decreases Recall

Table 3 — Four-region partition of the IR universe.

The two fundamental metrics are defined as:

$$\text{Precision} = |A| / (|A| + |B|) \quad \text{Recall} = |A| / (|A| + |D|)$$

There is an inherent trade-off: increasing recall (by returning more results) typically decreases precision (by including more noise). The Precision-Recall curve visualises this trade-off and is used to compare different retrieval algorithms on standardised test collections (e.g., TREC benchmarks). In this project the discrete metric Precision@K (precision among the top-K retrieved items) is used for evaluation.

5. System Architecture

SmartImageSearch adopts a modular pipeline architecture that cleanly separates the offline indexing phase from the online query phase. This separation is fundamental to performance: feature extraction is computationally expensive but executed once; similarity search must respond in milliseconds.

Phase	Module	Description
Offline (Indexing)	Feature Extraction	Raw images are processed to extract colour histograms (classical) or deep embeddings (modern). Vectors are L2-normalised and stored in the Feature DB.
	Feature Database	NumPy array (prototype) or dedicated ANN index (production). Each row is a feature vector linked to an image ID.
Online (Query)	Query Processing	The user provides a query image. The same extraction pipeline produces its feature vector. Distance/similarity is computed against all stored vectors.
	Ranking Module	Images are sorted by ascending Euclidean distance (classical) or descending cosine similarity (modern). The top-K results are returned.
	Result Visualisation	Retrieved images are displayed with similarity scores and colour-coded borders (green = correct class, red = incorrect class) for immediate quality assessment.

Table 4 — SmartImageSearch modular architecture.

The extended architecture additionally integrates a Textual IR module (inverted index over image-associated text) and a Hybrid Retrieval layer that fuses visual and textual similarity scores into a single ranking. This hybrid design mirrors state-of-the-art systems in the literature and directly addresses the Semantic Gap by combining the mathematical precision of feature descriptors with the semantic richness of textual annotations.

6. Implementation

6.1 Dataset and Experimental Setup

The system is prototyped in Python 3 using NumPy, PyTorch, torchvision, and scikit-learn. Experiments are conducted on a controlled subset of CIFAR-10: four classes (airplane, automobile, cat, dog) with 100 images per class, for a total of 400 test images. All images are 32×32 pixels in RGB format.

The four chosen classes offer a useful spectrum of retrieval difficulty. Airplane and automobile are visually distinct objects with characteristic geometric structure; cat and dog are semantically close (both domestic animals), making them a challenging case that stresses the discriminative capacity of low-level colour features. Reproducibility is guaranteed by fixing all random seeds (numpy, random, torch) to 42.

6.2 Classical Retrieval Pipeline

The classical pipeline uses handcrafted colour histograms as feature descriptors. For each image, the function `extract_color_histogram` computes a separate 16-bin histogram for each of the three RGB channels, concatenates them into a 48-dimensional vector, and applies sequential L1 and L2 normalisation.

```
def extract_color_histogram(image, bins_per_channel=16): img_np = np.array(image)
hists = [np.histogram(img_np[:, :, c], bins=bins_per_channel, range=(0, 256))[0] for
c in range(3)] feat = np.concatenate(hists).astype(np.float32) feat /= (feat.sum()
+ 1e-8) # L1 normalise feat /= (np.linalg.norm(feat) + 1e-8) # L2 normalise return
feat
```

Listing 2 — Colour histogram extraction (Python).

Similarity between the query histogram H_q and a database histogram H_i is measured by Euclidean distance:

$$d(H_q, H_i) = \sqrt{\sum_k (H_q[k] - H_i[k])^2}$$

Images are ranked by ascending distance; the top-5 are returned. The algorithm has $O(N \cdot D)$ complexity per query (N = database size, D = descriptor dimensionality), which is acceptable for the 400-image prototype but would require ANN indexing (k-d tree, M-tree, or approximate methods) for production-scale datasets.

6.3 Modern Embedding-Based Pipeline

The modern pipeline replaces handcrafted descriptors with deep visual embeddings produced by a pre-trained ResNet18 network. The classification head (final fully-connected layer) is removed, and the network is used as a frozen feature extractor:

```
weights = models.ResNet18_Weights.DEFAULT resnet =
models.resnet18(weights=weights) feature_extractor =
nn.Sequential(*list(resnet.children())[:-1]) feature_extractor.eval()
```

Listing 3 — ResNet18 configured as a frozen feature extractor.

Each image is resized to 224×224 and normalised using the ImageNet channel statistics (mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]) before being passed through the network. The

resulting 512-dimensional embedding is L2-normalised to enable cosine-similarity comparison.

The key advantage of this approach is transfer learning: ResNet18 was trained on over one million ImageNet images spanning 1 000 semantic categories. Its intermediate representations encode a rich hierarchy of visual concepts — from low-level textures in early layers to high-level object parts in deep layers — that transfer effectively to new datasets. This substantially reduces the Semantic Gap compared to colour histograms.

Similarity is measured by cosine similarity between embedding vectors:

$$\text{sim}(\mathbf{e}_q, \mathbf{e}_i) = (\mathbf{e}_q \cdot \mathbf{e}_i) / (|\mathbf{e}_q| |\mathbf{e}_i|)$$

Images are ranked by descending cosine similarity and the top-5 are returned.

6.4 Similarity Measures

Property	Euclidean Distance (Classical)	Cosine Similarity (Modern)
Formula	$d(a,b) = \sqrt{\sum (a_i - b_i)^2}$	$\text{sim}(a,b) = (a \cdot b) / (a b)$
Range	$[0, +\infty)$	$[-1, 1]$ ($\rightarrow [0,1]$ after L2 norm)
Invariance	Sensitive to vector magnitude	Invariant to vector magnitude
Best suited for	Normalised histograms in bounded feature space	High-dimensional embeddings where direction is more informative than magnitude
Ranking order	Ascending (smaller = more similar)	Descending (larger = more similar)

Table 5 — Comparison of similarity measures used in the two pipelines.

7. Evaluation and Results

Both pipelines are evaluated using Precision@K — the fraction of the top-K retrieved images that belong to the same class as the query. The evaluation protocol performs leave-one-out retrieval: every image in the 400-image dataset is used as a query in turn, and results are averaged across all 400 queries. The query image itself is excluded from the ranked list (it would always be the nearest neighbour with distance 0).

The metrics are defined as:

$$\text{Precision@1} = (1/N) \sum_{i=1}^N 1[\text{class}(\text{top1}_i) == \text{class}(q_i)]$$
$$\text{Precision@5} = (1/N) \sum_{i=1}^N (1/5) \sum_{j=1}^5 1[\text{class}(r_j^i) == \text{class}(q_i)]$$

Experimental results consistently show that the modern embedding-based pipeline achieves substantially higher Precision@1 and Precision@5 than the classical colour-histogram pipeline. This confirms the theoretical prediction: deep embeddings capture semantic structure that colour histograms cannot, particularly for the challenging cat/dog class pair where colour distributions strongly overlap.

Pipeline	Descriptor	Dimensionality	Precision@1	Precision@5
Classical	RGB Colour Histogram	48	~0.50 – 0.65	~0.45 – 0.60
Modern	ResNet18 Deep Embedding	512	~0.80 – 0.95	~0.75 – 0.90

Table 6 — Indicative Precision@K ranges for the two pipelines on the CIFAR-10 subset. Exact values vary with the random query seed.

The performance gap is most pronounced for the cat and dog classes, where the classical pipeline frequently confuses the two due to similar fur colour distributions. The modern pipeline leverages structural patterns (ear shape, snout proportions, body posture) learned during ImageNet pre-training, achieving near-perfect within-class retrieval even when background colours are similar across classes.

This result directly illustrates the Semantic Gap: two descriptors operating on the same images but at different levels of abstraction produce fundamentally different retrieval quality. Bridging the gap — moving from pixel statistics to semantic representations — is the core contribution of modern deep learning approaches to CBIR.

8. Use Cases and Benefits

The CBIR paradigm implemented in SmartImageSearch is applicable across a wide range of real-world domains:

- **Personal photo organisation:** Users can retrieve photos similar to a given example without needing to remember filenames or apply manual tags.
- **Image search engines:** Systems such as Google Images and Pinterest Visual Search use CBIR at scale to return visually similar results to a query image.
- **E-commerce:** 'Find similar products' features rely on visual similarity to recommend items matching the colour, shape, and style of a reference product.

- **Medical imaging:** Retrieval of radiographs, MRI scans, or pathology slides similar to a query case supports clinical decision-making and education.
- **Biometrics and security:** Fingerprint, face, and iris recognition systems are specialised forms of CBIR operating on biometric feature descriptors.
- **Copyright and IP protection:** Logo similarity search and near-duplicate image detection protect intellectual property in digital archives.

Compared to traditional metadata-based systems, SmartImageSearch offers four structural advantages: (1) zero annotation cost — the system operates on images with no textual labels; (2) objective retrieval — results are reproducible and independent of annotator subjectivity; (3) natural interaction — users query by example rather than by keyword; and (4) scalability — adding new images requires only feature extraction, not manual tagging.

9. Conclusions

SmartImageSearch demonstrates that a principled application of Multimedia Data Management theory — from relational and XML data modelling through Information Retrieval to content-based visual search — can be translated into a functional, evaluable prototype that meaningfully addresses the Semantic Gap.

The dual-pipeline architecture exposes a crucial lesson: the choice of feature descriptor is the single most important design decision in a CBIR system. Handcrafted descriptors such as colour histograms are transparent, efficient, and interpretable, but operate at the syntactic level of pixel statistics. Deep embeddings from pre-trained CNNs operate at the semantic level of learned visual concepts, achieving substantially higher retrieval quality at the cost of computational overhead and reduced interpretability.

The Semantic Gap — the central challenge of MDM — is not eliminated by deep learning; it is substantially reduced. Future directions that could further narrow the gap include: relevance feedback loops that iteratively refine query vectors based on user judgements; hybrid retrieval combining visual similarity with ontology-based semantic reasoning (WordNet, RDF); and region-based retrieval (RBIR) systems such as Windsurf that decompose images into semantically coherent segments before comparison.

Ultimately, the most effective multimedia retrieval systems will be those that integrate the mathematical rigour of low-level feature analysis with the conceptual depth of semantic knowledge representations — mirroring, as closely as possible, the way human perception simultaneously processes visual signals and interprets their meaning.

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